INCH-POUND
MIL-DTL-16377/72
2 August 1996

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; BALLAST, FOR ONE 17-WATT T-12 FLUORESCENT LAMP

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-16377 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

All dimensions in inches.
For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

AMSC N/A FSC 6210
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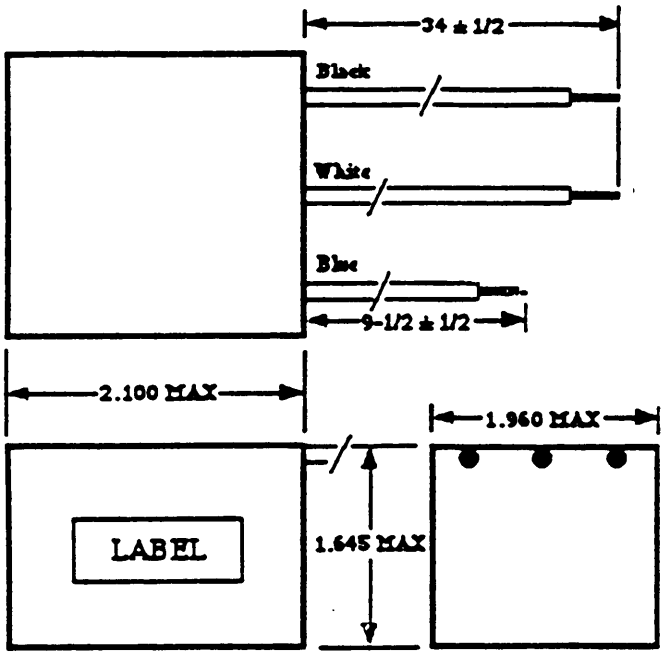
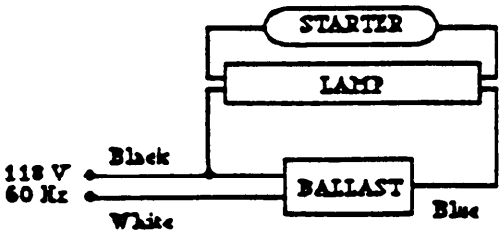


Leadwires to be #18 AWG, Stranded, UL Approved. Ends to be Stripped 3/8 ± 1/16. Temperature Requirements: 105°C (Thermoplastic) for Part No. 001; 150°C (Silicone Impregnated Fiberglass, SF-2) for No. 002.



LABEL: TYPE I

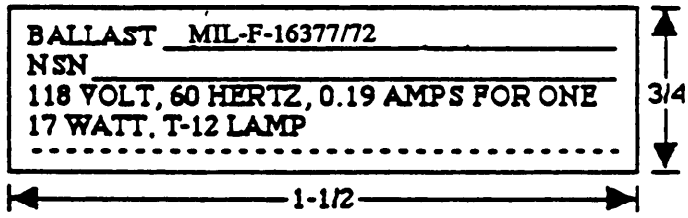


FIGURE 1. Ballast for one 17-Watt T-12 fluorescent lamp.

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TABLE I. Ballast rating and general information.

Military part no. M16377/72-	Rating				Efficiency	Ballast factor	Remarks
	Volt	Amperes	Hertz	Power factor			
001	118	0.19	60	≥90%	>80%	92.5%	Standard
002	118	0.19	60	≥90%	>80%	92.5%	Heat resistant

REQUIREMENTS:

DESIGN:

Dimension and configurations: See figure 1.
Assembly: The ballast shall consist of a laminated iron core and coil, integrally assembled and conforming to the dimensions and requirements shown herein.
Impregnating: Ballast circuitry to be housed in thermoset plastic potting shell and to be potted with epoxy.
Markings: Ballast shall conform to marking requirements and color coding of leads as specified in the applicable paragraph of American National Standards Institute (ANSI) C82.1, "Ballasts for Fluorescent Lamps - Specifications."
Label: Label shall be in accordance with type I except as specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-DTL-16377 and as follows:

First article inspection: See table II.
Conformance:
Comparison inspection: Same as the first article inspection (see table II for test procedures).
Inspection of product for delivery:
Examination: See MIL-DTL-16377.
Examination for preparation of delivery: See MIL-DTL-16377.
Conformance tests:

(a) Power output (except ratio peak current to r.m.s. current).
(b) Dielectric withstanding voltage.
(c) Insulation resistance.
(d) Voltage drop across ballast.
(e) Current harmonic distortion.
(f) Ballast efficiency.
(g) Ballast power factor.

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GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table I.

TABLE II. First article and comparison inspection.

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	3/
Performance (a) Power output (b) Regulation (c) Noise (d) Voltage drop across Ballast (e) Ballast efficiency (f) Ballast power factor	1/	2/	3/
Dielectric withstanding voltage	3.6.15	4.8.2	3/
Insulation resistance	3.6.16	4.8.3	3/
Current harmonic distortion	DOD-STD-1399, section 300 para 3.3.6.2	6.2.7	3/

1/ The performance of the ballasts shall be in conformance with the requirements listed in the applicable paragraph of ANSI C82.1, "Ballasts for Fluorescent Lamps - Specifications.", except that the voltage drop across the ballast shall be not less than 84 volts.

2/ Test procedure for performance shall be as follows:
- (a) Performance tests shall be made in accordance with ANSI C82.2, "Fluorescent Lamp Ballasts - Methods of Measurement."
- (b) Fluorescent lamp reference ballasts shall be in conformance with ANSI C82.3, "Reference Ballasts for Fluorescent Lamps."
- (c) Lamp shall conform with approved ANSI C78.2, "Dimensional and Electrical Characteristics of Fluorescent Lamps - Preheat-Start Types."
- (d) Voltage across starter terminals and across ballast, when supplying a reference lamp, shall be measured.

3/ Sample lot size for inspection shall consist of five ballasts.

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Reference ballast characteristics: The specifications described in table III ensure that the ballasts conform to requirements to operate the F17T12CW lamp, PPD 802-6215499.

TABLE III. Reference ballast characteristics for field light output. 1/

Input voltage	Lamp current	Impedance	Power factor	Volt-amps
118V	0.327A	276.9 ohms	0.071	29.6 V-A

1/ Test conditions (77°F ambient).

Preparing activity:
Navy - SH
(Project 6210-N661-13)

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